

SOKCHO, Korea, Republic of

WMO: 470900

Lat: 38.2508N

Lon: 128.5647E

Elev: 64

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.8	20.4	-14.0	2.6	21.4	-9.9	3.2	24.3	21.2	30.2	19.1	31.0	7.1	290	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	88.6	75.7	86.1	74.6	83.6	73.6	78.6	84.4	77.2	83.1	75.8	81.5	6.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.0	141.0	82.2	75.5	133.9	80.7	74.0	127.1	79.6	42.3	84.3	40.8	83.2	39.5	81.6	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	15.4	13.1	11.1	93.3	4.1	2.5	8.1	95.0	5.6	96.5	3.3	97.8	0.3	99.6
			7.7	79.6	3.8	1.9	5.0	81.0	2.8	82.2	0.7	83.2	-2.0	84.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	32.2	35.3	43.2	53.4	61.6	67.6	74.5	75.9	68.1	59.5	47.9	36.6	(6)
(7)		DBStd	16.07	6.26	6.35	7.19	7.10	6.73	5.05	5.89	4.78	4.25	5.23	7.07	7.03	(7)
(8)		HDD50	1783	550	411	236	41	1	0	0	0	5	121	418		(8)
(9)		HDD65	4646	1015	831	676	356	152	27	1	0	15	179	512	882	(9)
(10)		CDD50	3521	0	1	26	143	360	527	760	802	544	298	59	1	(10)
(11)		CDD65	907	0	0	1	7	45	103	297	337	109	8	0	0	(11)
(12)		CDH74	5737	0	0	1	38	241	461	2193	2451	344	8	0	0	(12)
(13)		CDH80	1530	0	0	0	5	33	81	711	655	45	0	0	0	(13)
(14)	Wind	WSAvg	5.4	5.9	5.8	6.1	6.4	5.7	4.6	4.4	4.3	4.6	5.0	5.7	5.8	(14)
(15)	Precipitation	PrecAvg	52.2	2.1	2.4	2.2	2.9	3.1	4.1	8.6	10.9	8.1	3.5	2.5	1.8	(15)
(16)		PrecMax	79.2	6.5	8.2	4.8	10.0	6.4	11.4	19.8	23.7	25.6	13.5	6.3	8.9	(16)
(17)		PrecMin	36.5	0.0	0.1	0.3	0.0	0.4	0.6	2.0	2.2	0.3	0.1	0.0	0.0	(17)
(18)		PrecStd	9.6	2.0	2.0	1.2	2.5	1.8	2.8	4.0	5.6	6.1	3.4	2.0	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.6	56.0	70.1	78.8	83.8	86.2	92.4	92.1	84.6	74.6	67.9	56.4	(19)
(20)		MCWB	41.3	44.1	52.6	58.6	61.3	66.9	76.6	77.9	72.4	60.7	55.0	46.1		(20)
(21)		2%	DB	46.6	51.6	62.9	73.1	79.7	82.2	89.2	88.5	80.6	71.9	64.4	52.7	(21)
(22)		MCWB	38.1	41.3	48.3	55.0	60.2	66.8	75.9	76.8	70.4	60.7	53.5	43.9		(22)
(23)		5%	DB	43.7	48.1	58.6	68.8	76.5	79.1	86.8	86.0	77.6	69.8	61.6	49.4	(23)
(24)		MCWB	36.1	39.4	46.3	52.6	59.0	66.1	75.1	75.8	69.5	60.6	51.9	41.3		(24)
(25)		10%	DB	41.1	45.0	54.9	65.2	73.3	76.2	84.2	83.6	75.4	68.0	58.9	46.6	(25)
(26)		MCWB	34.0	37.1	44.2	51.2	58.0	65.9	74.3	75.4	68.5	59.4	50.3	39.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	46.5	54.3	63.0	66.0	72.6	80.1	80.9	75.5	66.9	58.4	48.7	(27)
(28)		MCDWB	48.2	53.6	66.2	73.1	76.1	78.7	86.5	87.3	81.4	70.2	63.3	53.9		(28)
(29)		2%	WB	39.6	42.6	50.6	57.7	63.5	70.8	78.3	79.2	72.7	64.6	55.8	45.1	(29)
(30)		MCDWB	44.8	49.2	60.1	67.4	73.5	76.4	85.1	84.8	77.6	68.8	61.5	51.2		(30)
(31)		5%	WB	37.3	40.3	47.6	55.7	61.9	69.5	76.7	77.9	71.1	62.9	53.8	42.1	(31)
(32)		MCDWB	42.1	46.4	55.9	64.5	70.6	74.6	84.1	83.1	75.9	67.4	59.8	48.1		(32)
(33)		10%	WB	35.1	38.2	45.4	53.5	60.5	68.1	75.4	76.7	69.7	61.2	51.5	39.9	(33)
(34)		MCDWB	39.8	43.7	53.2	61.7	68.5	73.1	82.0	81.7	74.1	66.0	57.6	45.6		(34)
(35)	Mean Daily Temperature Range		MDBR	11.7	12.0	13.1	13.6	12.8	9.8	8.5	9.1	10.9	12.7	12.1	11.9	(35)
(36)		5% DB	MCDBR	13.5	15.1	17.3	19.2	18.7	15.5	13.0	12.2	12.9	14.2	14.5	13.6	(36)
(37)		MCWBR	10.4	10.4	10.3	9.9	8.7	6.4	5.5	5.9	7.7	9.3	10.5	10.5		(37)
(38)		5% WB	MCDBR	11.7	14.1	16.2	16.2	15.0	11.0	11.3	10.1	10.7	11.6	12.9	12.7	(38)
(39)		MCWBR	9.7	10.7	10.9	10.0	8.3	5.8	5.7	6.0	7.2	8.8	10.5	10.6		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.410	0.471	0.559	0.611	0.616	0.664	0.631	0.555	0.486	0.484	0.448	0.403	(40)	
(41)		taud	1.994	1.870	1.730	1.675	1.713	1.678	1.809	2.003	2.119	1.989	2.007	2.041	(41)	
(42)		Ebn at Noon	228	229	222	220	223	213	218	233	241	224	213	220	(42)	
(43)		Edh at Noon	41	54	68	76	75	77	67	54	45	47	41	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	780	1022	1294	1556	1722	1639	1359	1344	1244	1074	759	689	(44)	
(45)		RadStd	72	105	126	129	172	131	186	167	148	93	103	66	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	-3.73	N/A	N/A	N/A	N/A	N/A	+184	+112	
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+178	+94	(47)

Nomenclature: See separate page